

Customer Behaviour Analysis

About the Dataset

Type: Retail / E-commerce Dataset

Nature: Structured (Tabular)

Level: Customer Transaction Level

Data Mix: Numerical → Age, Purchase Amount, Previous Purchases

Categorical → Category, Payment Method, Season

This dataset is used for **customer behavior analysis + business decision-making**

Domain Background

In the **e-commerce industry**, companies track every customer interaction:

What customers buy

How often do they buy

How much do they spend

Whether they respond to discounts

Why This Data Is Important

This data helps businesses:

Identify **high-value customers**

Understand **purchase patterns**

Optimize **discount strategies**

Improve **marketing campaigns**

Increase **customer lifetime value (CLV)**

Business Problem

How can a retail company understand customer purchasing behavior to increase revenue, improve customer retention, and optimize marketing strategies?

Breaking Down the Problem

1. Low Customer Retention

Customers are not purchasing frequently

Need to understand:

Who are frequent buyers?

Who are one-time buyers?

2. Ineffective Marketing Campaigns

Discounts and promo codes may not be effective

Questions:

Do discounts increase purchases?

Which customers respond to promo codes?

3. Product Demand Understanding

Which category performs best?

Which products generate the highest revenue?

4. Payment Behavior Insights

Which payment methods are preferred?

Are digital payments increasing?

5. Seasonal Demand Variation

Does season impact sales?

Which season drives maximum purchases?

6. Customer Segmentation Problem

Need to identify:

High-value customers

Frequent buyers

Discount-sensitive customers

Data Dictionary

Column Name	Description
Customer ID	Unique identifier for each customer
Age	Age of the customer
Gender	Gender of the customer (Male/Female/Other)
Item Purchased	Specific product purchased
Category	Product category (Clothing, Electronics, etc.)
Purchase Amount	Amount spent by the customer
Location	Customer's geographic location
Size	Product size (S, M, L, etc.)
Color	Product color
Season	Season of purchase (Winter, Summer, etc.)
Review Rating	Customer rating for the product
Subscription Status	Whether customer is subscribed (Yes/No), Subscription Program
Shipping Type	Type of shipping selected
Discount Applied	Whether discount was applied
Previous Purchases	Number of past purchases

Payment Method	Mode of payment
Frequency of Purchases	Purchase frequency (Weekly, Monthly, etc.)

EDA-(Exploratory Data Analysis using Python)

```
import pandas as pd***
```

```
df=pd.read_csv("/Users/milanmanoj/Documents/My Projects/Customer Behaviour Analysis/customer_shopping_behavior.csv")
```

```
df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Previous Purchases	Payment Method
0	2701	22	Female	T-shirt	Clothing	68.0	California	XL	Olive	Winter	3.2	No	Standard	No	36.0	Cash
1	521	51	Male	Sunglasses	Accessories	84.0	South Carolina	Free Size	White	Spring	3.9	Yes	Free Shipping	Yes	20.0	Debit Card
2	3157	18	Female	Shirt	Clothing	50.0	Montana	M	Black	Winter	3.1	No	2-Day Shipping	No	18.0	Cash
3	1687	22	Male	Gloves	Accessories	75.0	Illinois	Free Size	Red	Fall	4.2	No	Store Pickup	No	25.0	Cash
4	2929	40	Female	Jewelry	Accessories	80.0	Alabama	Free Size	Yellow	Spring	3.6	No	Store Pickup	No	17.0	Credit Card

Initial Data Exploration

Viewed data using:

```
df.head()
```

```
df.tail()
```

Checked dataset summary:

```
df.describe(include='all')
```

```
df.info()
```

Checked missing values:

```
df.isnull().sum()
```

To understand:

Data structure

Data types

Missing values

This step helps identify data quality issues early

Data Consistency Check (Category Correction)

Checked unique values in:

Item Purchased

Category

Found inconsistencies between Item Purchased and Category

Created a mapping dictionary

Categories were inconsistent or incorrect
Ensures:
Accurate grouping
Correct business insights

Handling Missing Values (Size Column)

Handled missing values based on category:
Electronics → "Not Applicable."
Accessories → "Not Applicable."
Footwear → "Free Size."
Clothing → Filled with mode

Different categories require different business logic
Example:

Electronics don't have size → Not Applicable
Clothing → size is important → use mode

Handling Missing Values (Review Rating)

Checked:
Mean, median, mode
Filled missing values using:
Rounded values

Instead of the global average, the product-level mean is used
This keeps ratings realistic and meaningful

Handling Missing Values (Purchase Amount)

Replaced with the mean value.

Maintains product-level pricing consistency
Avoids distortion in revenue analysis

Handling Missing Values (Previous Purchases)

Replaced with the 0 (zero) value.

Missing values imply no previous purchase
Important for:
Customer segmentation
Loyalty analysis

Duplicate Handling

Checked duplicates:
Identified duplicates using Customer ID
Removed duplicates:

Ensures:
Each customer appears only once
No double-counting in analysis

Column Standardization

Converted column names:
df.columns = df.columns.str.replace(" ", "_")
df.columns = df.columns.str.lower()
Renamed column:

```
df.columns=df.columns.str.replace(" ", "_")  
df.columns=df.columns.str.lower()
```

Makes column names:
Clean
SQL-friendly
Easy to use in Tableau

Data Export to SQL Server

Installed required libraries:
pip install pyodbc sqlalchemy
Connected to SQL Server
Exported data:

To move cleaned data into SQL for further analysis
Enables:
Query-based insights
Integration with BI tools.

1. Which category generates the highest revenue?

category	highest_reven...
Electronics	486366.96
Accessories	194866.87
Clothing	183188.39
Footwear	111236.2
Outerwear	18524

2. Are discounts actually increasing purchase value?

discount_appli...	total_revenue
No	502241.64
Yes	491940.78

3. What is the total revenue generated by male and female customers?

gender	total_revenue
Male	390964.53
Female	352060.29
Other	251157.6

4. Which customer used a discount but still spent more than the average purchase amount?

customer_id	discount_appli...	purchase_amount
4442	Yes	201.93
4558	Yes	208.02
4032	Yes	213.14
4046	Yes	216.55
4581	Yes	230.19
5000	Yes	238.66
4361	Yes	238.88

5. Which are the top/bottom 5 products with the highest average review rating?

item_purchas...	avg_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.79

item_purchas...	avg_rating
Phone	2.94
Headphones	3.08
Watch	3.1
Bag	3.17
Laptop	3.19

6. Average purchase: Standard vs Express shipping

shipping_ty...	order_placed	avg_purchase	total_amou...
Express	1199	358.12	429380.98
Standard	1201	340.57	409020.45

7. Do subscribed customers spend more? Compare average speed and total revenue between subscribers and non-subscribers.

subscription_sta...	users	average_spend	total_spe...
No	3421	165.72	566937.02
Yes	1579	270.58	427245.4

8. Top 5 products with highest discount usage %

item_purchas...	total_no_of_times_s...	total_no_of_times_sold_when_discount_ap...	discount_percenta...
Laptop	143	74	51.7483
Phone	171	88	51.4620
Headphones	166	85	51.2048
Hat	154	77	50.0000
Watch	157	78	49.6815

9. Segment customers into new, returning, and loyal based on their total number of previous purchases, and show the count of each segment.

customer_segment	customer_count
Loyal Customer	3085
Returning Customer	1354
New Customer	561

10. What are the top 3 most purchased products within each category ?

category	item_purchas...	most_purchas...	rnk
Accessories	Jewelry	171	1
Accessories	Bag	164	2
Accessories	Sunglasses	161	3
Accessories	Belt	161	3
Clothing	Shirt	315	1
Clothing	Pants	171	2
Clothing	Blouse	171	2

11. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

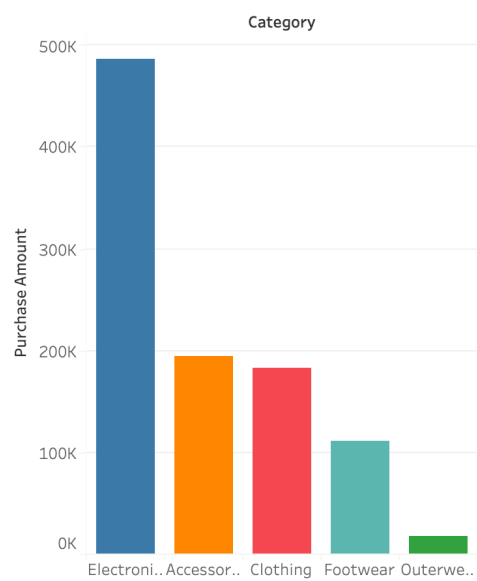
customer_type	customer_count	subscription_sta...	percent
Normal Buyers	658	No	62.84623
Normal Buyers	389	Yes	37.15377
Repeat Buyers	2763	No	69.89628
Repeat Buyers	1190	Yes	30.10372

12. What is the revenue contribution of each age group?

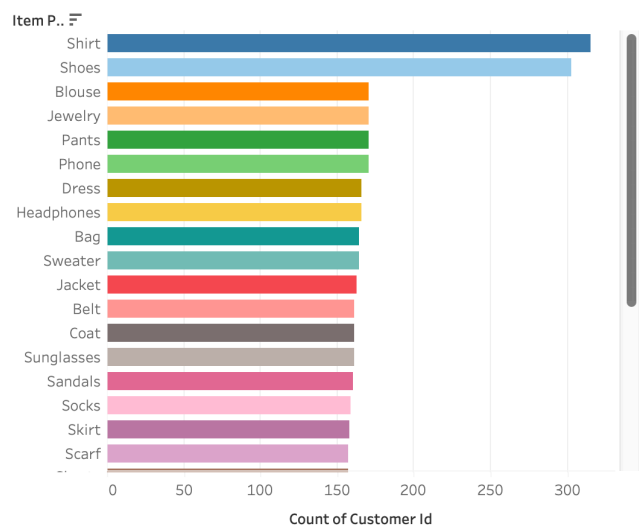
age_group	total_revenue
51+	399345.32
36-50	266706.7
26-35	177421.93
18-25	150708.48

Tableau Dashboard

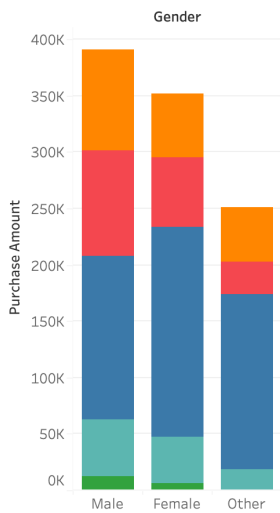
Category Sales



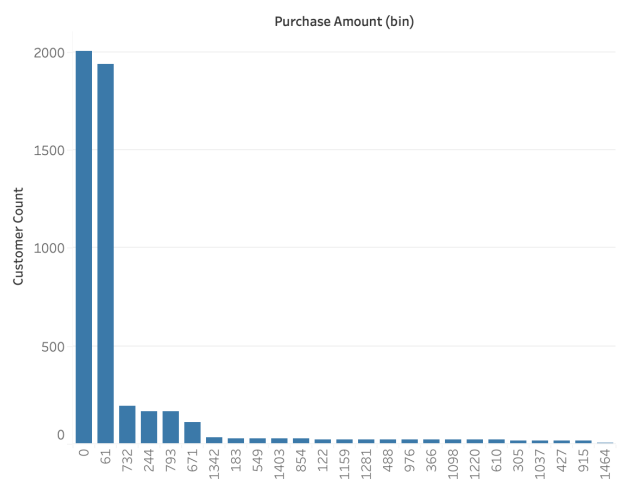
Item Popularity



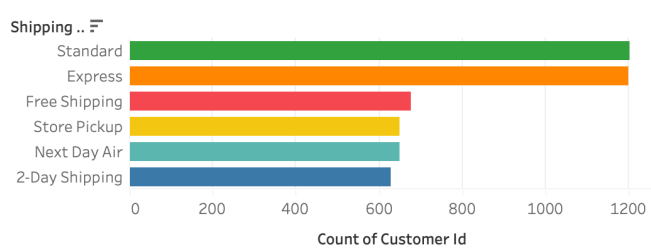
Gender vs Sales



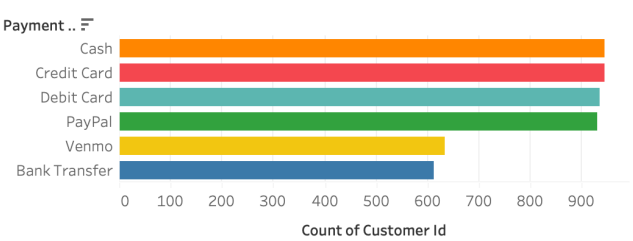
Purchase Distribution



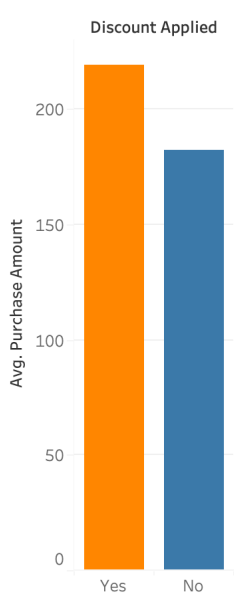
Shipping Analysis



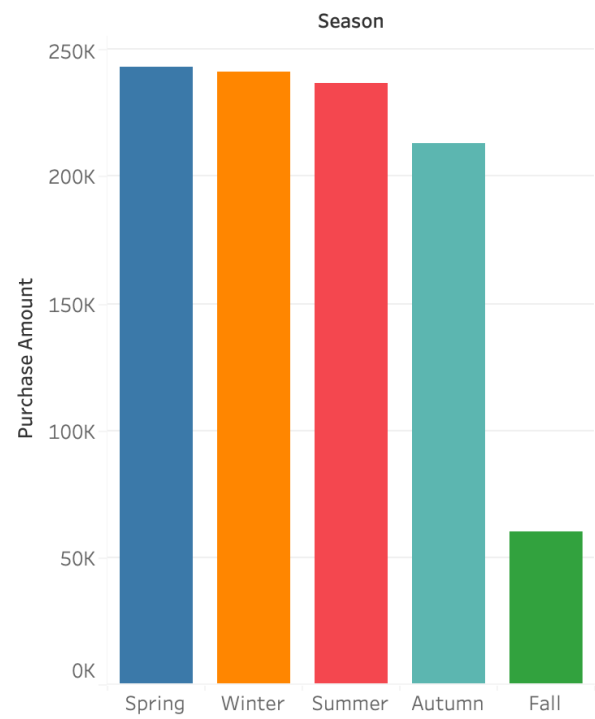
Payment Method



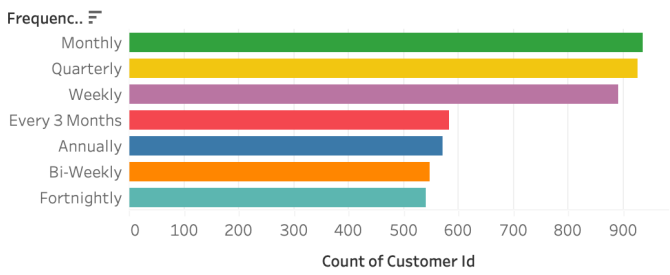
Discount Impact



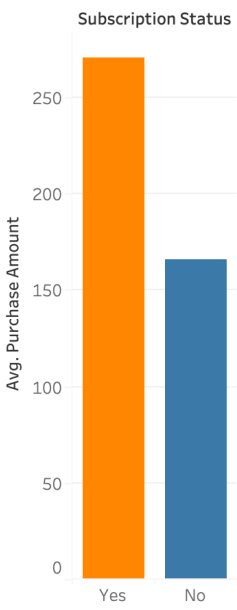
Seasonal Sales



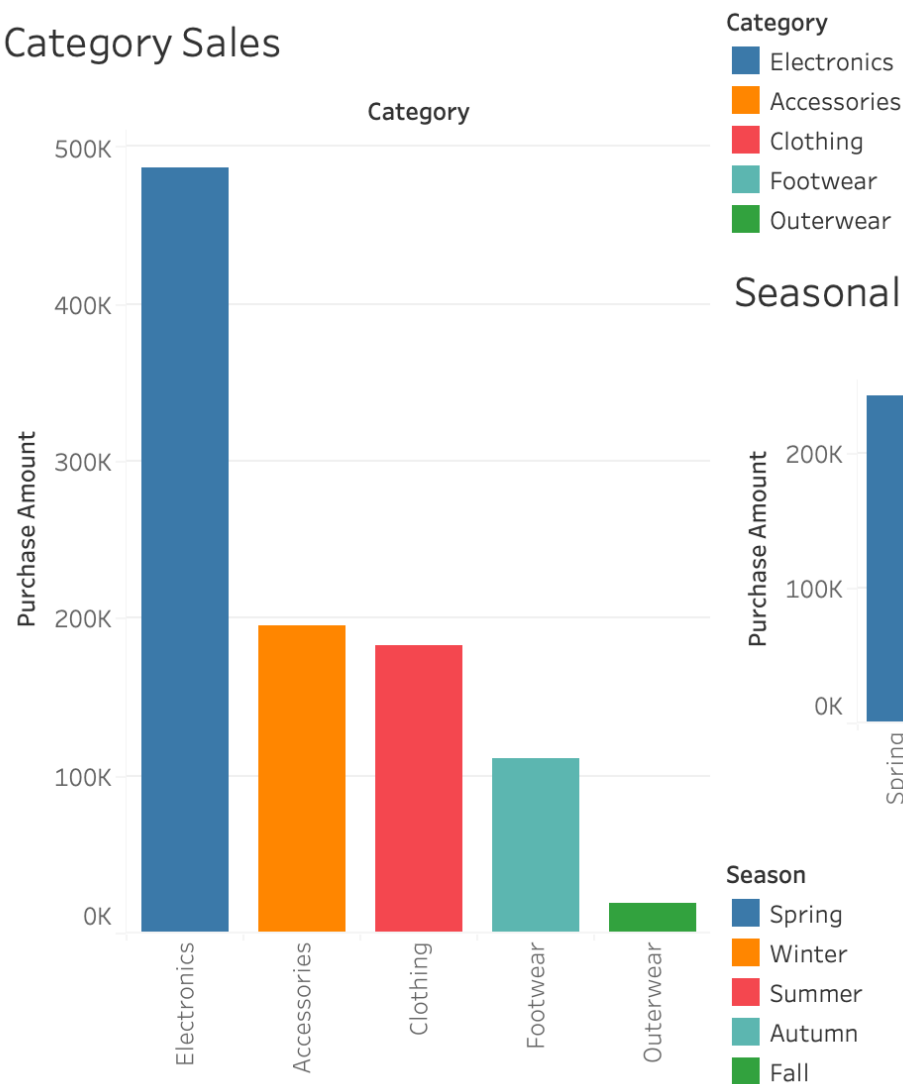
Purchase Frequency



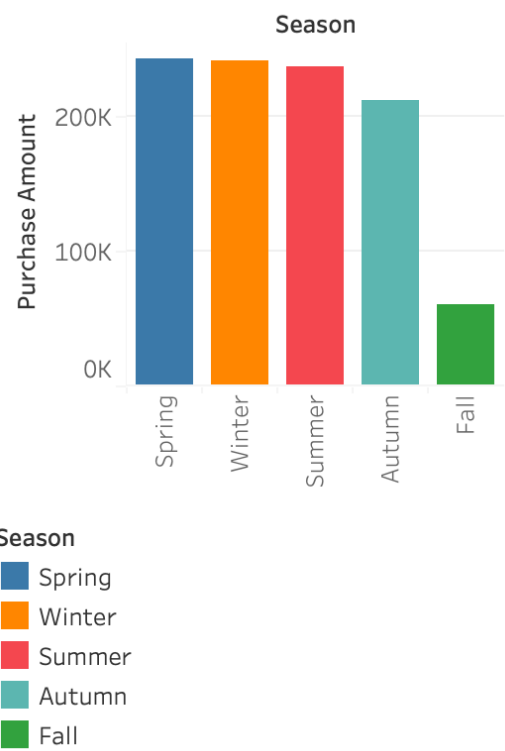
Subscription Analysis



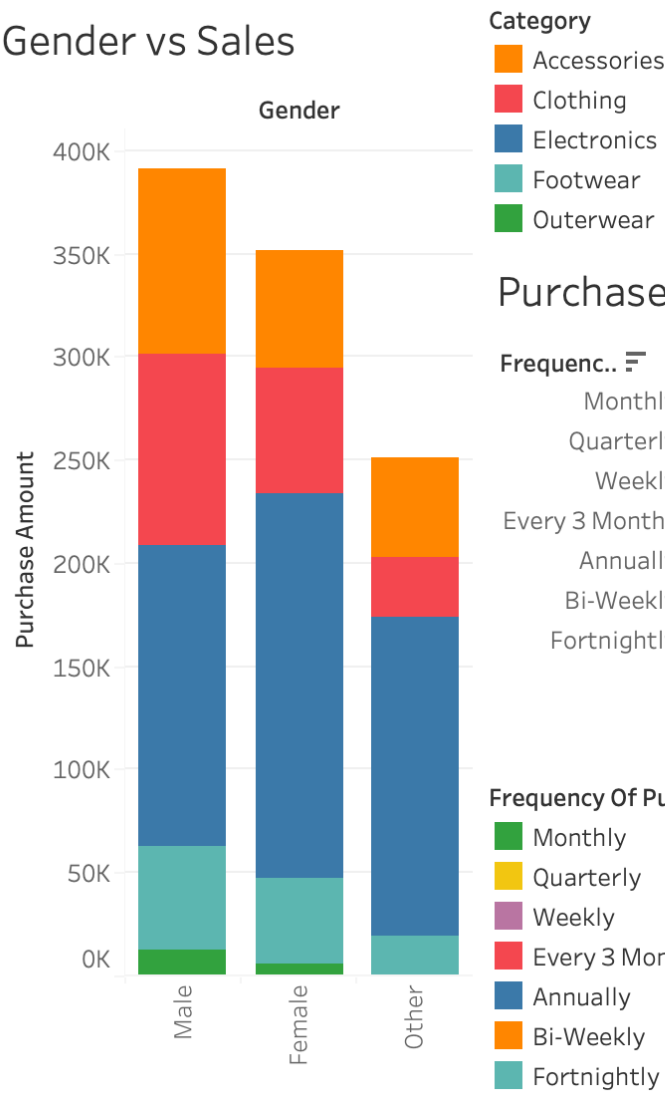
Category Sales



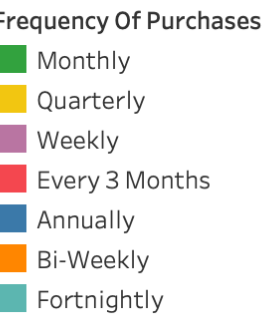
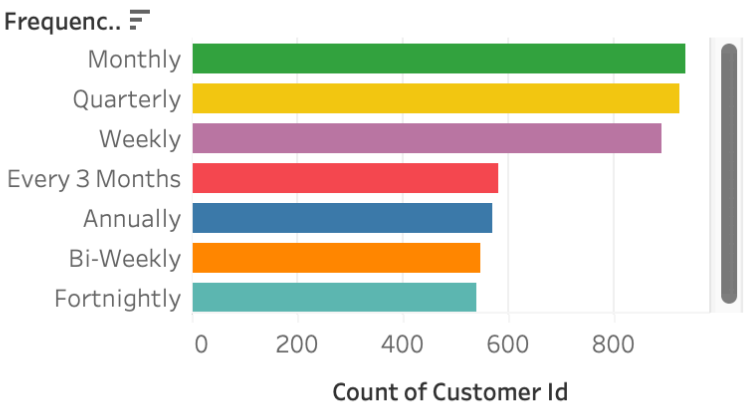
Seasonal Sales



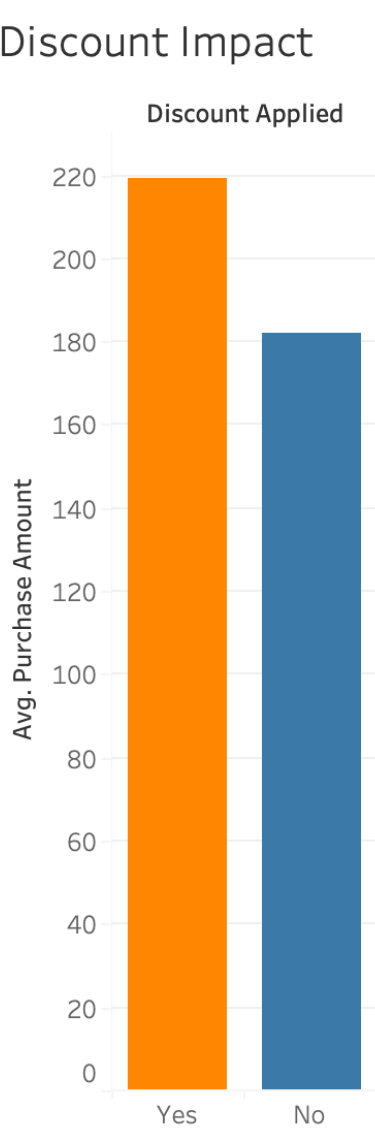
Gender vs Sales



Purchase Frequency



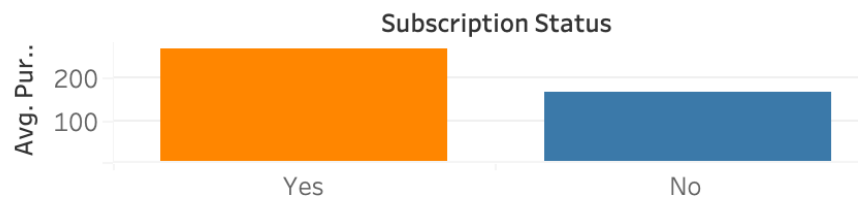
Discount Impact



Discount Applied

- Yes
- No

Subscription Analysis



Subscription Status

- Yes
- No

Shipping Analysis

Shipping Type

- Standard
- Express
- Free Shipping
- Store Pickup
- Next Day Air
- 2-Day Shipping